DE–3641 | **13**

DISTANCE EDUCATION

B.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2012.

DIGITAL COMPUTER FUNDAMENTALS

(2007 onwards)

Time : Three hours | Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Convert the following binary numbers to decimal

   (i) 111101

   (ii) 111.11.

   (b) Convert the following decimal numbers to octal.

   (i) 99

   (ii) 49.625

2. (a) Describe in detail about the BCD, Excess-3 and Gray codes.

   (b) Using 9's and 10's complement method perform the following operations

   (i) 75 – 50

   (ii) 50 – 75.

3. (a) Draw the logic circuit for

   $$F(A, B, C) = \overline{A}B + A\overline{B}\,\overline{C} + AB\overline{C} + ABC.$$

   (b) Explain the basic laws of Boolean algebra with truth tables.

4. (a) Simplify using Karnaugh map.

$Y = F(A, B, C, D) = \Sigma(2, 3, 12, 13, 14, 15)$.

(b) Draw the circuit for an 8-input multiplexer and explain.

5. (a) What is a full subtractor? How is a full subtractor built using two half subtractors?

(b) Draw a neat diagram for a 4-bit adder circuit and explain it.

6. (a) Explain the working of clocked Rs flip-flop using NOR gates.

(b) What is racing in JK flip-flop? How it is solved in JK master-slave flip-flop?

7. (a) What are the different uses of a shift registers.

(b) Explain the working of BCD counter.

8. (a) Compare the advantages and disadvantages of static RAM and Dynamic RAM.

(b) Discuss the functions of Input-output units.

DE–3642 | **14**

DISTANCE EDUCATION

B.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2012.

DATA STRUCTURES USING C

(2007 onwards)

Time : Three hours | Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 =100)

1. (a) List out and explain the library function in C language.

   (b) Define the following terms and give one example

      (i) Identifiers

      (ii) Constants

      (iii) Expression

      (iv) Keywords.

2. (a) Discuss the different type of Loop Statements in C language.

   (b) Write a program in C language to find the factorial of n numbers.

3. (a) How to define and accessing the function in C language.

   (b) Write a program in C language to calculate Fibonacci sequence of n number.

4. (a) Explain the different type of Arrays in C language.

   (b) Write a program in C language to concatenate of two Strings.

5. (a) How do you pass structure to a function? Explain with an example program.

   (b) How do you process the Data file? Explain with an example.

6. (a) Write an algorithm in Stack operation.

   (b) Discuss about the Binary Tree Traversal.

7. (a) Explain the create, insert and delete algorithm of double linked list.

   (b) Briefly explain Threaded Binary Tree.

8. (a) Write an Huffman algorithm.

   (b) How do you represent list as Binary Trees.

DE–3643 **15**

DISTANCE EDUCATION

B.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2012.

DISCRETE MATHEMATICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

(5 × 20 = 100)

1. (a) Show that $\neg(P \wedge Q) \rightarrow (\neg P \vee (\neg P \vee Q)) \Leftrightarrow (\neg P \vee Q)$.

   (b) Obtain the principal disjunctive normal form of $P \rightarrow ((P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P))$.

2. (a) Prove that

   (i) $(A - B) - C = (A - C) - (B - C)$

   (ii) $A - (A \cap B) = A - B$.

   (b) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are one-to-one and onto mappings. Prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

3. (a) (i) Give an example of a relation which is neither reflexive nor irreflexive.

   (ii) Give an example of a relation which is both symmetric and antisymmetric.

   (b) Let $X = \{1, 2, ...7\}$ and $R = \{< x, y > / x - y$ is divisible by 3}. Show that R is an equivalence relation. Draw the graph of R.

4. (a) Define subsemigroup and submonoid. Prove that for any commutative monoid $(m,*)$, the set of idempotent elements of m forms a submonoid.

(b) Define :

(i) homomorphism mapping

(ii) epimorphism

(iii) endomorphism

(iv) automorphim

(v) isomorphism mapping.

5. (a) In any group $(G,*)$, prove that $(a*b)^{-1} = b^{-1} * a^{-1}$ for any $a,b \in G$.

(b) State and prove Lagranges theorem.

6. (a) Prove that every subgroup of a cyclic group is cyclic.

(b) Prove that

(i) Every subgroup of an abelian group is a normal subgroup.

(ii) Let $f : G \to G'$ be a homomorphism, prove that $\ker(f)$ is a normal subgroup of G.

7. (a) Let $f : G \to G'$ be a group homomorphism. Prove that

(i) $f(e) = e'$, where e, e' are the identity elements of G and G' respectively.

(ii) For any $a \in G, f(a^{-1}) = f(a)^{-1}$.

(b) Prove that a non empty subset S of a group G is a subgroup of $(G,*)$ if and only if for any $a,b \in S$, $a * b^{-1} \in s$.

8. (a) Let $f : A \to B$ and $g : B \to C$ be functions. Prove that

(i) if both f and g are one-to-one, $g \circ f$ is also one-to-one.

(ii) if both f and g are onto, $g \circ f$ is also onto.

(b) Let $f : R \to R$ and $g : R \to R$ where R is the set of real numbers, be given by $f(x) = x^2 - 2$ and $g(x) = x + 4$. Find $f \circ g$ and $g \circ f$. State whether these functions are injective, surjective and bijective.

DISTANCE EDUCATION

B.Sc. (IT) DEGREE EXAMINATION, MAY 2012.

COMPUTER LAB – I (C AND DATA STRUCTURES)

(2007 onwards)

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by Lot System.

Each subdivision carries 50 marks.

1. (a) Write a C program to input a basic pay and calculate Gross pay and net pay

Income :

HRA = 15% of Basic pay

DA = 90% of Basic pay

Gross pay = Basic pay + HRA + DA

Deductions :

PF = 20% of Basic pay

LIC = 200

Deductions = PF + LIC

Net pay = Gross pay – Deductions.

(b) Write a C program to implement push and pop operations on stack.

2. (a) Write a C program to find the minimum and maximum in a given set of data.

(b) Write a C program to implement insert and delete operations on queue.

---------- Cut here ----------

3. (a) Write a program to convert the given number into words.

(b) Write a C program to sort the given elements using insertion sort.

---------- Cut here ----------

4. (a) Write a C program to sort the given set of names in alphabetical order.

(b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

5. (a) Write a C program to concatenate two given strings and find the length.

(b) Write a C program to evaluate the given mathematical expression using stack.

---------- Cut here ----------

6. (a) Write a C program to find the factorial of a given positive number using function. Otherwise display suitable message.

(b) Write a C program to implement insert and delete operations on Linked List.

7. (a) Write a menu driven program in C to find an element using Linear and binary search methods.

(b) There are 10 students in a class. Their names and marks in three different subjects are given. If a student takes more than 40 marks in each subject, then he is declared 'PASS'. Otherwise 'FAIL'. Write a C program to do the above.

---------- Cut here ----------

8. (a) Write a C program that receives the data such as age and name of person to check the eligibility for voting. Take the condition that if a person is more than 18 years old he is eligible to vote, else display the number of years, he has to wait for voting.

(b) Write a C program to sort the elements using selection sort.

DE–3644 **23**

DISTANCE EDUCATION

B.Sc. (IT)/B.Sc. (IT)/(Lateral) DEGREE EXAMINATION, MAY 2012.

INTERNET PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is a network? Explain in detail about network topologies. (10)

   (b) What is TCP/IP? Explain about the importance and terminology of TCP/IP. (10)

2. (a) Explain in detail about the Domian Name system concepts and functions. (10)

   (b) What is HTTP? Explain about methods and client requests in HTTP. (10)

3. (a) Discuss about the usage of internet at home in detail. (10)

   (b) Describe the usage of internet at office. (10)

4. (a) Explain about the basic concepts and principles of object oriented programming. (10)

(b) List out various data types in JAVA with example. (10)

5. (a) Explain about classes and objects in JAVA with example. (10)

(b) Write about constructors and object reference variable in JAVA. (10)

6. (a) Explain about mail transfer protocol in detail. (10)

(b) Describe about File transfer protocol in detail. (10)

7. (a) Write about variable declaration and Dynamic initialization in JAVA. (10)

(b) Explain about type conversion and casting in JAVA. (10)

8. (a) What is inheritance? Explain any two types of inheritance in JAVA. (10)

(b) What is overloading? Explain overloading methods in JAVA.With an example. (10)

DE–3645 **24**

DISTANCE EDUCATION

B.Sc.(IT)/B.Sc(IT) (Lateral) DEGREE EXAMINATION, MAY 2012.

ANALYSIS AND DESIGN OF INFORMATION SYSTEMS

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$ marks)

1. (a) What are the Types of Systems? Explain them.

   (b) Explain about the Role of Systems Analyst.

2. (a) Explain about the Feasibility Study.

   (b) Describe the System Analysis.

3. (a) List out the Design Methodologies and explain them.

   (b) What is Form Design? Explain the Characteristics of Good Form Design.

4. (a) What are the types of Database? Explain them.

   (b) Explain about HIPO and IPO Charts.

5. (a) Explain about System Maintenance.

   (b) List out the Conversion Methods and explain them.

6. (a) Explain about System Development Life Cycle.

(b) Explain the Multifaced Role of a System Analyst.

7. (a) What are the Steps in Feasibility Analysis? Explain them.

(b) Explain about Process and Stages of System Design.

8. (a) Describe the Quality Assurance.

(b) Explain the Procedure of Hardware and Software Selection.

DE–3646 **25**

DISTANCE EDUCATION

B.Sc. (IT)/B.Sc. (IT) (Lateral) DEGREE EXAMINATION, MAY 2012.

APPLICATION PROGRAMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Explain how can you

   (i) Create a folder, rename and delete the folder. (4)

   (ii) Change the background colour of the desktop (3)

   (iii) Create short cut for an application. (3)

   (b) Discuss the menu commands available in windows MS-office. (10)

2. (a) Discuss the concept of mail merge.

   (b) Create a invitation of your college function in MS-word by using formatting text.

3. (a) Create a excel sheet for your class internal mark statement. Then use total, average and rank in the sheet by using mathematical functions.

   (b) What is the use of formula bar? Give any five formula available and its usages.

4. (a) How can you create a presentation using the design template?

(b) Write the procedures to create and edit powerpoint presentation.

5. (a) Write the procedures to create and save a table in MS-Access.

(b) Create a table for your bank details using MS-Access.

6. (a) How can you insert table in MS-word document?

(b) Write the commands and its purpose for formatting text.

7. (a) What are the formatting commands available in MS-Excel? Explain it in detail.

(b) Create a sheet for customer details of particular shop, and sort the details based on their name.

8. (a) How can you combine excel data and charts with word documents?

(b) What is the use of handouts in power point? Explain it with your example.

DE–4362 **26**

DISTANCE EDUCATION

B.Sc. (IT)/B.Sc. (IT) (Lateral) DEGREE EXAMINATION, MAY 2012.

COMPUTER LAB – II (JAVA PROGRAMMING LAB)

(2007 onwards)

Time : Three hours Maximum : 100 marks

ONE question should be selected by the Examiner and given to each candidate by lot.

Each subdivision carries 50 marks.

1. (a) Write a Java program for sorting a given list of names in ascending order using command line arguments.

   (b) Write a Java program to display the contents of file along with a line number before each line.

---------- Cut here ----------

2. (a) Write a Java program to multiply the given two matrices.

   (b) Write a Java program to count the number of character, lines and works in a text file.

---------- Cut here ----------

3. (a) Write a Java program to illustrate the overloading and overriding methods.

   (b) Write a Java program to read and write the data using random access file.

4. (a) Write a Java program to illustrate the single and multilevel inheritance.

(b) Write a Java program that reads file name from the user and display the contents of file.

---------- Cut here ----------

5. (a) Write a Java program to illustrate the implementation of abstract class.

(b) Write a Java program to create and manipulate the Text Area, canvas, scroll bars, Frames, Menus and Dialog boxes.

---------- Cut here ----------

6. (a) Write a Java program to create a package.

(b) Write a Java program to count the number of words and characters in a text.

---------- Cut here ----------

7. (a) Write a Java program to create multiple threads.

(b) Write a Java program to draw various polygons.

---------- Cut here ----------

8. (a) Write a Java program to implement producer/consumer problem using synchronization.

(b) Write a Java program to illustrate the use of files and streams.

DE–4363 **27**

DISTANCE EDUCATION

B.Sc. (IT)/B.Sc. (IT) (Lateral) DEGREE EXAMINATION, MAY 2012.

COMPUTER LAB III — APPLICATION PROGRAMS LAB

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer ONE question from each Unit, chosen by Examiner.

MS-Word : 25
Excel : 35
Access : 40

UNIT I — MS-WORD

1. Prepare your resume. Use table, bullets and different colour features.

———————————— cut here ————————————

2. Prepare a news report using two columns, insert a picture in the first column and make the text flow around it.

———————————— cut here ————————————

3. Prepare an invitation for a function to be conducted in your institution. Use different text orientation and pictures to make it attractive.

———————————— cut here ————————————

4. Apply for the post of programmer to different companies using mail merge.

UNIT II — MS EXCEL

1. Draw Line, Bar and Pie charts for the data given below :

XYZ Ltd.

Year	Sales	Cost	Profit
2001	1000	40	600

Add data for 10 years. Provide titles, legends, grids and data labels.

---------- cut here ----------

2. Create a data table to create a ready reckoner table for a commercial bank that contains simple interest for Rs. 1,000 with varying period and interest rates.

---------- cut here ----------

3. Create a worksheet with the following columns :

Salesman no., name, city, product name, amount.

Add three records for five different salesmen who have carried out sales of different products in different cities. Find :

(a) Citywise and salesmanwise total sales

(b) Grand total sales.

---------- cut here ----------

4. Create two worksheets containing day to day house hold expenses for the months January and February 2010, with the following columns :

ITEM	AMOUNT
Stationary	50.00

(a) Add 10 different items for two months in two different worksheets.

(b) Consolidate both the months data and find the total expenses on each item.

(c) Find the total expenses for two months.

UNIT III — MS ACCESS

1. Create Employee table with the fields :

   EMPNO, NAME, AGE, SEX, STREET, CITY, PIN, SALARY.

   (a) Add data for 20 employees.

   (b) Show all the records in the table for the city "CHENNAI" by filtering.

   (c) Write a query to display all female employee whose age is in the range 20 to 30.

---------- cut here ----------

2. Create STUDENT table with the fields :
   REGNO, NAME, $M1$, $M2$, $M3$
   Create ADDRESS table with the fields :
   REGNO, STREET, CITY, PIN.

   (a) Write a query to display REGNO, NAME and total of $M1$, $M2$ and $M3$.

   (b) Write a query to display REGNO, NAME, STREET, CITY, PIN and total of $M1$, $M2$ and $M3$.

---------- cut here ----------

3. Create INVENTORY table with the fields :

   ITEMNO, NAME, QUANTITY ON HAND.

   (a) Create a form in custom format.

   (b) Create a query REORDER to show all items.

   (c) Create a macro that executes REORDER query automatically.

---------- cut here ----------

4. Create two tables SALES 1 and SALES 2. Join them to produce a third table SALES 3.

DE–3647 **31**

DISTANCE EDUCATION

B.Sc. (IT)/ B.Sc. (IT) (Lateral) DEGREE EXAMINATION, MAY 2012.

COMPUTER NETWORKS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. $(5 \times 20 = 100)$

All questions carry equal marks.

1. (a) Explain about Network Hardware.

   (b) What are the Transmission Media and explain.

2. (a) Explain about Elementary Data Link protocol.

   (b) What is Protocol Verification and explain?

3. (a) Explain briefly about Congestion Control Algorithms.

   (b) Discuss about Network Layer Design Issues.

4. (a) Explain briefly about TCP Protocol.

   (b) What are the Performance Issues? Explain briefly them.

5. (a) Explain about Cryptography with an illustration.

(b) Describe the Electronic mail and Data Compression.

6. (a) Explain about TCP/IP Reference Model.

(b) List out the Communication Satellites and explain.

7. (a) Explain about Multiple Access Protocols.

(b) Describe the DataLink layer Design Issues.

8. (a) Explain the Network layer in the Internet.

(b) List out the Elements of Transport Protocols.

DE–3648 **32**

DISTANCE EDUCATION

B.Sc. (IT)/B.Sc. (IT)/(Lateral) DEGREE EXAMINATION, MAY 2012.

VISUAL PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about format and Icons of a window. (10)

   (b) Describe the different window messages. (10)

2. (a) Write about control panel and print manager of windows. (10)

   (b) What do you mean by OLE? Explain Picture attributes of windows. (10)

3. (a) Write a window program to create own brush and display lines with it. (10)

   (b) Write a note on Input Box with an example. (10)

4. (a) How do you create menus in VB? Explain Menu Editor. (10)

   (b) Write a note on Frames with example program. (10)

5. (a) Explain the following :

(i) Branching with IF. (5)

(ii) SELECT.....CASE. (5)

(b) Define function. Describe the concept of recursive function. (10)

6. (a) What are Control Arrays? Explain with the help of a suitable example. (10)

(b) Write the properties and methods of Scroll Bars. (10)

7. (a) Explain ReDim and preserve keywords with suitable examples. (10)

(b) Give short notes on Forms. (10)

8. (a) What do you understand by event-driven programming? List and Explain about some of the events supported by Forms and Command button objects. (10)

(b) Explain the different ways of arranging the windows on MDI form. (10)

DISTANCE EDUCATION

B.Sc. (IT)/B.Sc. (IT)/(Lateral) DEGREE EXAMINATION, MAY 2012.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the developments in communication technology.

   (b) Write a note on ethics of Information Technology.

2. (a) Write a note on presentation graphics software.

   (b) Discuss the importance of Database Softwares.

3. (a) Explain about Internet Web browsers.

   (b) What is specialized software? Explain.

4. (a) What do you mean by work group computing? Explain.

   (b) Explain about online information services.

5. (a) What is Electronic data interchange? Explain.

   (b) Discuss cyber ethics in detail.

6. (a) Discuss the criteria for crating secondary storage devices.

   (b) Write a note on optical disks.

7. (a) Explain five management systems in detail.

(b) Describe the different types of database organisation.

8. (a) Explain about the six phases of system analysis and design.

(b) Discuss Internet programming in detail.

DE–3650 **34**

DISTANCE EDUCATION

B.Sc.(I.T)/ B.Sc.(I.T) (Lateral)DEGREE
EXAMINATION, MAY 2012.

MOBILE COMPUTING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss 3G and 4G technologies in detail.

   (b) Write a note on Network signaling.

2. (a) Discuss the applications of Mobile Ip.

   (b) Explain GPRS services in detail.

3. (a) Explain WAP protocol application environment in detail.

   (b) Write a note on wireless application protocol gate ways.

4. (a) Discuss implementing WAP protocol strategy.

   (b) Write a note on WAP client Software and Hardware.

5. (a) Write a note on markup languages.

   (b) Discuss about writing and formatting text.

6. (a) Explain about displaying images and Tables in detail.

   (b) Write a note on using Variables and Acquiring User Input.

7. (a) Write a note on WML script.

   (b) Discuss about WML script control structures.

8. (a) Write a note on WML script Events.

   (b) Explain various usabilities of WML script.

DE–3651 **35**

DISTANCE EDUCATION

B.Sc.(I.T)/ B.Sc.(I.T)(Lateral) DEGREE EXAMINATION, MAY 2012.

RDBMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the advantages of DBMS. (8)

   (b) Explain the database schema. (12)

2. (a) Explain the basic concepts of relational model. (8)

   (b) What do you mean by data control facilities? Explain. (12)

3. (a) Write a note on client/ server system. (8)

   (b) Write a note on Oracle users. (6)

   (c) Explain the benefits of Oracle. (6)

4. (a) Explain DML with examples. (12)

   (b) Explain about NULL values in detail. (8)

5. (a) Create a database for a college system. (8)

(b) Explain how can you define a table and drop a table. (12)

6. (a) Describe the data control facilities. (10)

(b) Discuss about different database user interface. (10)

7. (a) Explain the architecture of Oracle. (10)

(b) Explain the Oracle logical structure. (10)

8. (a) Explain table join with examples. (12)

(b) Explain union with examples. (8)

DE–4364 **36**

DISTANCE EDUCATION

B.Sc. (Information Tech.)/B.Sc. (IT) (Lateral) DEGREE EXAMINATION, MAY 2012.

LAB IV — VISUAL BASIC AND ORACLE

(2007 onwards)

Time : Three hours Maximum : 100 marks

(1) Examiner has to select TWO questions ONE from VB and another from ORACLE.

(2) Each question carries 50 marks.

(3) Questions are to be given to the candidates by lot system.

(4) Write algorithm and draw flowchart if necessary.

VB

1. Write and test a VB program to compute and print either the SUM or the PRODUCT of the first *N* natural numbers. Use option button.

---------- Cut here ----------

2. Write a VB program to do temperature conversion C to F and F to C at user's choice using label, text, enter key.

---------- Cut here ----------

3. Write and test a VB program to display the day of the given date and convert upper case from lower case to given word.

4. Write and test a VB program to read in principal, number of years and rate of interest through Input boxes, compute and print Simple Interest, Compound Interest through labels.

---------- Cut here ----------

5. Write a VB program to calculate Simple Interest (or) Compound Interest using the command buttons namely input, calculate, display.

---------- Cut here ----------

ORACLE

6. Create a table ADDRESS with the following structure :

Field name	Type	Width	Decimal
REG NO	Numeric	8	
NAME	Character	20	
MARK1	Numeric	3	
MARK2	Numeric	3	
MARK3	Numeric	3	
TOTAL	Numeric	3	
RESULT	Character	4	

(a) Add 5 records.

(b) Display the structure of the table.

(c) Fill TOTAL field with sum of MARK1, MARK2, MARK3.

(d) Fill the RESULT field with 'PASS' if TOTAL >= 150 otherwise "FAIL".

7. Create table PAY with the following structure.

Field name	Type	Width	Decimal
EMP NO	Numeric	8	
NAME	Character	25	
DOJ	Date	8	
BPAY	Numeric	8	2
DA	Numeric	8	2
HRA	Numeric	3	
DEDU	Numeric	3	
GPAY	Numeric	8	2
NPAY	Numeric	8	2

(a) Add 5 records (Fill all fields except DA, HRA, DEDU, GPAY, NPAY)

(b) Fill DA with 75% of PAY for all employees.

(c) Fill HRA with 300 for all employees.

(d) Fill DEDU with 750 for all employees.

(e) Fill GPAY with sum of BPAY, DA, HRA.

(f) Fill NPAY with GPAY – DEDU.

(g) Display EMPNO, NAME, BPAY, DEDU, GPAY, NPAY alone.

8. Create table PERSONAL with following structure.

Field name	Type	Width	Decimal
SNO	Numeric	8	
NAME	Character	20	
CITY	Character	20	
PIN	Numeric	6	
PHONE	Numeric	6	

(a) Add 5 records.

(b) Display the contents of the table.

(c) Display all names.

(d) Display all names without duplicate.

(e) Display all names in upper case, lower case and initial capital letters.

(f) Display all names and length of names.

(g) Display name, city in which only beginning portion of name is known.

9. Create table SALES with the following structure.

Field name	Type	Width	Decimal
SALE NO	Numeric	5	
SALE NAME	Character	20	
SALE AMT	Numeric	8	2
COMM	Numeric	6	2
SALE DATE	Date		

(a) Add 5 records.

(b) Replace COMM with the following conditions :

If sales amount is less than 1,000 there is no commission.

If sales amount is greater than (or) equal to 1,000 but less than or equal to 5,000, commission is 5% sales amount.

If sales amount is greater than 5,000, commission 10%.

(c) Display all the records.

(d) Display the records having commission < 1000

(e) Display the records having commission > 1000

(f) Display the average sales and average commission.

10. Create table BLOOD with the following structure.

Field name	Type	Width	Decimal
DNO	Numeric	3	
DNAME	Character	20	
DOB	Date		
DAGE	Numeric	2	
DADD1	Character	15	
DADD2	Character	15	
DCITY	Character	15	
DPIN	Numeric	6	
DBLOOD	Character	3	
DSEX	Character	1	

(a) Add 5 records.

(b) Display all the records.

(c) Display the information of donors in the age group 20 to 25 using between option.

(d) Display the information of donors having date of birth between 01-JAN-55 and 31-DEC-75.

(e) Display the information of donors having any of blood group from the set of blood groups in using IN option.

(f) Display the information of female donors with age between 20 and 25 using between option.

DISTANCE EDUCATION

B.Sc. (IT)/B.Sc. (IT) (Lateral) DEGREE EXAMINATION, MAY 2012.

LAB V — NET WORKING

(2007 onwards)

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by lot system.

1. Write a java program to add two numbers using RMI client, server technique.

---------- Cut here ----------

2. Write a java program to find factorial of given number using FACTORIAL INTERFACE technique.

---------- Cut here ----------

3. Write a java program find factorial of given number using RMI client, server technique.

---------- Cut here ----------

4. Write a java program to setup a chat server and chat client.

---------- Cut here ----------

5. Write a java program to display to users and their IP address logged in the network.

6. Write a java program to read a file from the remote system.

---------- Cut here ----------

7. Write a java program to setup time server and time client.

---------- Cut here ----------

8. Write a java program to setup echo server and echo client.